

M1.1: Bus Node Modeling

Theory: Foundation of load flow. Buses are classified into three types based on boundary conditions: PQ bus (known load injection P & Q), PV bus (known generation P & voltage magnitude V, controlling Q to maintain V), and Slack bus (reference bus, known V & phase $\theta = 0$, balances system losses).

Details: Variables to solve: $P_{Qbus}(V_i, \theta_i)$; $PVbus(Q_i, \theta_i)$. Real/reactive power balance equations must converge.

Trade-off: Generator Q limits constrain PV buses. If reactive power Q violates its boundary during iteration, the PV bus temporarily degrades into a PQ bus using the limiting value.

M1.2: Pi equivalent Line Model

Theory: Models short/medium lines. Consists of a series impedance $Z = R + jX$ and shunt susceptances $Y/2 = jB/2$ split equally at both terminals.

Details: Series branch admittance $Y_{series} = 1/Z = G_{series} + jB_{series}$. Shunt charging models reactive support.

Trade-off: For long transmission lines (>150km), lumped-parameter models fail, requiring hyperbolic correction: $Z' = Z \sinh(\gamma l) / \gamma$ to prevent voltage profile distortions.

M1.3: Ybus Admittance Matrix

Theory: Describes grid connectivity. Grid size $N \times N$. Diagonal element Y_{ii} is the sum of admittances of branches connected to bus i . Y_{ij} is the negative branch admittance between i and j .

Details: Formulations: $Y_{ii} = y_{i0} + \sum y_{ik}$ and $Y_{ij} = -y_{ij}$.

Trade-off: High-voltage grids are sparse (>99% zero entries). Storing Y_{bus} in CSR formats and employing sparse solvers (e.g., KLU, UMFPACK) is mandatory for memory efficiency.

M1.4: Generator Limit Cost Curve

Theory: Defines active power production costs and secure operating zones. Operational generation cost is modeled via quadratic polynomial: $C_i(P_i) = a_i P_i^2 + b_i P_i + c_i$.

Details: Operational bounds: $P_{min} \leq P \leq P_{max}$ and $Q_{min}(P) \leq Q \leq Q_{max}(P)$, bounded by thermal limits and excitation rotor angle.

Trade-off: Neglecting the $P-Q$ limit coupling results in unrealistic OPF dispatches that can cause equipment stress.

M1.5: ZIP Load Model

Theory: Captures static load dependency on voltage magnitude. Comprises Constant Impedance (Z), Constant Current (I), and Constant Power (P) terms.

Details: Equations: $P = P_0 [a_1(V/V_0)^2 + a_2(V/V_0) + a_3]$ with $a_1 + a_2 + a_3 = 1$.

Trade-off: Simply modeling loads as constant power ($a_3 = 1$) during deep voltage drops causes severe numerical divergence.

M2.1: Newton-Raphson Method

Theory: Linearizes power flow equations via Taylor expansion, solving variables iteratively using derivatives of the Jacobian matrix.

Details: Step: $\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = J \begin{bmatrix} \Delta \theta \\ \Delta V/V \end{bmatrix}$. Updated variables: $\theta^{(k+1)} = \theta^{(k)} + \Delta \theta$.

Trade-off: Recomputing and factorizing the Jacobian at each iteration is computationally expensive.

M2.2: Fast Decoupled Power Flow

Theory: Simplifies Jacobian under the physical assumption that $X \gg R$, which decouples $P-\theta$ and $Q-V$ channels in high-voltage grids.

Details: Form: $\Delta P/V = B' \Delta \theta$ and $\Delta Q/V = B'' \Delta V$. Constant matrices B' and B'' are factorized once.

Trade-off: Extremely fast and robust for sparse systems, but less accurate than Newton-Raphson.

M2.3: DC Power Flow

Theory: A completely linearized formulation assuming flat voltages ($V=1.0$), negligible resistance ($R \approx 0$), and small angle differences.

Details: Linear relation: $P = B \theta$. Line power flow: $P_{ij} = b_{ij}(\theta_i - \theta_j)$.

Trade-off: Ignores reactive power, voltage variations, and non-linearities.

M2.4: Optimal Power Flow (OPF)

Theory: Minimizes generation costs or transmission losses subject to operational bounds.

Details: Form: $\min \sum C_i(P_i)$ subject to non-linear flow equality $Sh(V, \theta) = 0$ and inequality constraints ($V_{min} \leq V \leq V_{max}$, branch flow limits $S \leq S_{max}$).

Trade-off: A highly non-convex NP-hard problem. Numerical interior point methods require careful initialization to avoid getting stuck in local minima.

M2.5: Detailed Generator Model

Theory: Models electromotive forces and magnetic flux paths. Varies from 3rd-order (transient EMF) to 6th-order (subtransient double-axis) formulations.

Details: Tracks state variables: $\delta, \omega, e'_d, e'_q, e''_d, e''_q$.

Trade-off: Accurately simulates machine dynamics but increases parameter complexity, demand for data, and simulation time.

M3.1: AVR & Governor Controls

Theory: AVR controls field excitation to hold terminal voltage; Governor regulates mechanical input power to control speed and grid frequency.

Details: AVR model (e.g., IEEE DC1A): $T_A \dot{V}_d = -E_{fd} + K_A(V_{ref} - V_t)$. Governor: $T_g \dot{P}_{valve} = -P_{valve} + P_{set} - P_{ref}$.

Trade-off: Excessive AVR gain (K_A) introduces negative damping, leading to instability.

M3.2: DAE Integration Algorithms

Theory: Power dynamics form a rigid DAE set: $\dot{x} = f(x, y)$ (dynamic states), $g(x, y) = 0$ (algebraic network constraints).

Details: Solved via implicit methods like the Trapezoidal Rule: $x_{n+1} = x_n + \Delta t \cdot \frac{1}{2}(f(x_n, y_n) + f(x_{n+1}, y_{n+1}))$.

Trade-off: DAE systems are numerically stiff, requiring robust solvers.

M3.3: Network Interface Admittance

Theory: Bridges machine equations and the grid admittance equations by modeling generators as voltage sources behind transient reactances.

Details: Norton equivalent current: $I_{inj} = E' / (jX' + Z_{ext})$.

Trade-off: Augmented admittance: $I = (Y_{bus} + Y_{gen})V$. Rapid changes in generator states during near-faults can trigger numerical current spikes if time-step is too high.

M3.4: Small-Signal Stability

Theory: Evaluates robustness against minor load fluctuations by linearizing the DAE around the current operating point.

Details: Form: $\Delta \dot{x} = A \Delta x$. Eigenvalues $\lambda_i = \sigma_i \pm j\omega_i$ dictate stability. Damping ratio: $\zeta = \frac{-\sigma_i}{\sqrt{\sigma_i^2 + \omega_i^2}}$.

Trade-off: Damping ratios below 3% trigger low-frequency power oscillations. Ignoring multi-machine interactions can result in unpredicted inter-area oscillations.

M3.5: Equal Area Criterion (EAC)

Theory: Graphical analysis of transient stability for single-machine systems under large disturbances.

Details: Assesses rotor swing angle. Accelerating area $A_{acc} = \int (P_m - P_{e}^{fault}) d\delta$ must be compensated by decelerating area $A_{dec} = \int (P_m - P_{e}^{post}) d\delta$.

Trade-off: Strictly limited to SMIB systems. Multi-machine networks require full time-domain simulations.

M4.1: Critical Clearing Time (CCT)

Theory: Maximum duration a fault can persist without causing the generator to lose synchronism.

Details: Derived by integrating the swing equation: $\int_{\delta_{cct}}^{\delta_{max}} (P_m - P_e(\delta)) d\delta = 0$.

Trade-off: Low inertia renewable grids compress the CCT drastically, requiring fast sub-cycle relaying and circuit breaker clearing.

M4.2: Droop & AGC Controls

Theory: Two levels of frequency defense. Primary droop control regulates speed in seconds; Secondary AGC adjusts base generation setpoints in minutes.

Details: Droop equation: $P_{gen} - P_{set} = -\frac{1}{R_d} (f - f_0)$. AGC balances Area Control Error: $SACE = \Delta P_{interchange} + B \Delta f$.

M4.3: BESS SoC Modeling

Theory: Battery Energy Storage Systems act as voltage sources. The State of Charge (SoC) tracks remaining energy capacity.

Details: SoC integration: $SoC(t) = SoC(0) - \eta \int \frac{I_{batt}(\tau)}{Q_{nominal}} d\tau$. Terminal voltage: $V_{batt} = V_{oc}(SoC) - I_{batt} R_{int}$.

M4.4: Inverter Droop Control

Theory: Mimics physical generator behavior by establishing voltage and frequency references in grid-connected power electronics.

Details: Equations: $\omega = \omega_{ref} - m_p(P_{calc} - P_{ref})$ and $V = V_{ref} - n_q(Q_{calc} - Q_{ref})$.

Trade-off: Lacking physical rotor mass, inverter droop control lacks pass filter cutoff in measurements can induce high-frequency oscillations.

M5.1: Active Distribution Grid (ADN)

Theory: High penetration of single-phase solar generators forces modeling of unbalanced three-phase distribution topologies.

Details: Uses a 3×3 impedance matrix for branches. State estimation uses $\min \|z - h(x)\|^2 R^{-1} \|z - h(x)\|^2$.

Trade-off: Three-phase estimation increases complexity by 9x. Low telemetry reduces data volume.

M5.2: Virtual Synchronous Generator (VSG)

Theory: Implements swing equations in inverter controls to provide virtual inertia and damping to low-inertia grids.

Details: Control loops: $J \frac{d\omega}{dt} = T_m - T_e - D(\omega - \omega_0)$.

Trade-off: High virtual inertia improves frequency stability, but excessive transient current demands can trigger overcurrent shutdowns of the inverter IGBTs.

M5.3: Islanding & Black Start

Theory: Islanding detection isolates microgrids for safety; Black Start re-energizes a dead grid using local battery grids and virtual generators.

Details: Active frequency drift shifts phase during islanding to trigger trips. Black Start uses Grid-Forming inverters to establish voltage reference without grid presence.

Trade-off: Active detection blind spots (NDZ) occur when load and generation match. Cold-load transformer inrush currents can easily collapse black start inverters.

M5.4: Wind DFIG & PMSG LVRT

Theory: Ensures wind turbines remain connected during deep grid faults to avoid cascading generation loss.

Details: DFIG clamps rotor overcurrents via a Crowbar circuit, turning it into an induction machine. PMSG dissipates excess power in a DC Chopper to protect DC link caps.

Trade-off: Activating the DFIG Crowbar consumes significant reactive power from the grid, degrading local voltage recovery during the fault.

M5.5: Solar MPPT Algorithms

Theory: Continuously tracks maximum output point of PV arrays under changing solar irradiance.

Details: Perturb and Observe (P&O) steps voltage periodically to maximize power; Incremental Conductance (INC) solves $dI/dV = -I/V$.

Trade-off: P&O oscillates around peak and fails under fast weather transients. Partially shaded arrays exhibit multi-peak curves, requiring global algorithms (e.g., PSO).

M6.1: LMP Pricing Structure

Theory: Price of supplying the next MW of load at a specific node, reflecting marginal costs of congestion and losses.

Details: Composition: $LMP_i = \lambda_{system} + \mu_{congestion,i} + \eta_{loss,i}$. Congestion is calculated using Shift Factors (SF).

Trade-off: Grid congestion splits prices, creating congestion surpluses. Poor financial transmission rights (FTR) distribution.

M6.2: Inertia Dilution

Theory: Inverter-based resources do not contribute physical inertia, diluting system equivalent inertia as fossil plants are decommissioned.

Details: Rate of Change of Frequency (RoCoF) limit: $\frac{\Delta P}{\Delta t} \leq f_{0,2H_system}$.

Trade-off: High RoCoF triggers fast protection trips. Mitigating requires expensive synchronous condensers or reserving active headroom in wind/solar grids.

M6.3: Card 27 Fallback Title

Placeholder text for compiler robustness.

M6.4: Card 28 Fallback Title

Placeholder text for compiler robustness.

Zone T1: PowerSystems.jl Core APIs

System(), add_component!(), to_json(), solve_powerflow(), ResidualModel(), DynamicSimulator(), sim_inputs(), solve_opf()

Zone T2: Grid Faults & Protections

fault_clearing_time, critical_clearing_time, rotor_angle_instability, voltage_collapse, frequency_deviation, UFLS, UVLS, droop_control, AGC